UNIT V

Failure Theories of Composites

Micromechanics of failure | anisotropic strength theories | importance of shear strength | choice of criteria

I. Micromechanics of Failure of a Unidirectional Lamina

The strength of a lamina depends on **how the fibre, matrix and interface fail**. Micromechanics of failure explains the five basic strengths (X_t, X_c, Y_t, Y_c, S) from constituent properties and identifies the **failure mode** for each loading.

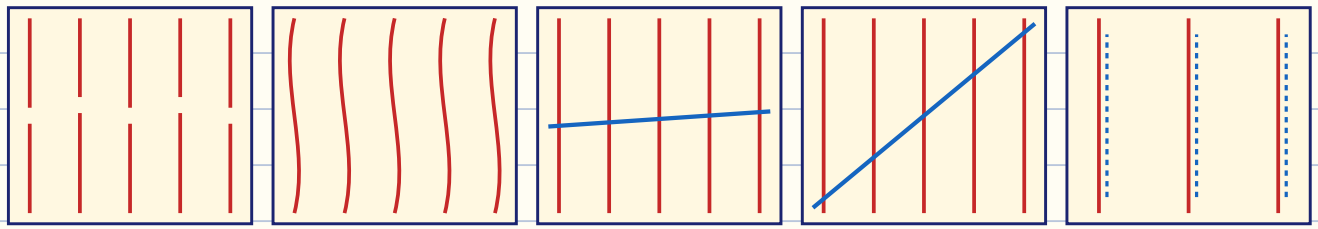


Fig. 1 Typical failure modes of a unidirectional lamina

Loading	Dominant failure mechanism	Controlled by
Longitudinal tension (X_t)	Fibre fracture (+ pull-out, matrix cracking, debonding)	Fibre strength, V_f
Longitudinal compression (X_c)	Fibre microbuckling / kink band, shear crushing	Matrix shear modulus, fibre alignment
Transverse tension (Y_t)	Matrix cracking, interface debonding, voids	Matrix & interface strength
Transverse compression (Y_c)	Matrix shear failure on 45° plane	Matrix strength
In-plane shear (S)	Matrix shear cracking, fibre-matrix interface shear	Matrix, interface

I.1 Longitudinal tensile strength Xt

Fibres are **brittle with scatter in strength** (flaws), so they break at random points at different loads. After a fibre breaks, load is transferred back to it through the matrix over an **ineffective length**, and neighbouring fibres see a stress concentration (Rosen's chain-of-bundles / Zweben models). Two cases decide the simple strength estimate ($\varepsilon_{fu} < \varepsilon_{mu}$, brittle fibre):

$$\sigma_{lu} = \sigma_{fu} V_f + \sigma'_m (1 - V_f) \text{ for } V_f > V_{crit}$$
$$\sigma_{lu} = \sigma_{mu}(1 - V_f) \text{ for } V_f < V_{min} \text{ (fibres only dilute the matrix)}$$
$$V_{min} = (\sigma_{mu} - \sigma'_m) / (\sigma_{fu} + \sigma_{mu} - \sigma'_m), \quad V_{crit} = (\sigma_{mu} - \sigma'_m) / (\sigma_{fu} - \sigma'_m)$$

σ_{fu} = fibre ultimate strength, σ_{mu} = matrix ultimate strength, σ'_m = matrix stress when the fibres fail ($\sigma'_m = E_m \varepsilon_{fu}$). In practice $X_t \approx 0.8$-0.9 of the rule of mixtures because of fibre misalignment and stress concentration.

Example

> $\sigma_{fu} = 2000$ MPa, $\sigma_{mu} = 60$ MPa, $\sigma'_m = 30$ MPa → $V_{min} = 30/2030 =$ **1.5 %**, $V_{crit} = 30/1970 =$ **1.5 %** : only above ≈ 1.5 % fibre the fibres begin to strengthen the matrix.

> At $V_f = 0.6$: $\sigma_{lu} = 2000(0.6) + 30(0.4) =$ **1212 MPa**.

1.2 Longitudinal compressive strength Xc

Under compression the fibres behave like columns on an elastic foundation (the matrix) and **microbuckle**. Rosen's analysis gives two modes:

Extension (in-phase) mode: $\sigma_c = 2V_f \sqrt{(V_f E_f E_m / 3(1 - V_f))}$

Shear (out-of-phase) mode: $\sigma_c = G_m / (1 - V_f)$

Example (glass/epoxy, $V_f = 0.6$, $E_f = 72$, $E_m = 3.5$, $G_m = 1.3$ GPa): extension mode = $2(0.6)\sqrt{(0.6 \times 72 \times 3.5/1.2)} =$ **13.5 GPa**; shear mode = $1.3/0.4 =$ **3.25 GPa** → the **shear mode governs**. Actual X_c (≈ 0.6-1.2 GPa) is far lower because of **fibre misalignment**, which gives kink-band formation at $\sigma_c \approx \tau_y/\varphi$ (φ = initial misalignment angle). Hence $X_c < X_t$ especially for aramid fibres.

1.3 Transverse tensile strength Yt

- Fibres are stiffer than the matrix, so the **matrix strain is magnified** between fibres: $\varepsilon_m = S_m \varepsilon_2$, where strain-magnification factor $S_m = 1 / [1 - V_f(1 - E_m/E_f)]$ (≈ 2.5 for $V_f = 0.6$ glass/epoxy).
- Failure starts at **interface debonding**, voids or matrix cracking: $Y_t \approx \sigma_{mu}/S_m$ or lower; so Y_t is only 3-5 % of X_t and is the **weakest** strength - the first failure in a laminate.
- Improved by good fibre-matrix adhesion (sizing), tougher matrix, lower void content.

1.4 Transverse compressive strength Yc

- Matrix compression crushes the matrix: failure is by **shear on planes at ≈ 45°-55°** to the load; fibres reinforce the matrix so $Y_c \approx 3$-$5 \times Y_t$.
- Interface crushing and fibre-matrix debonding add to the failure.

1.5 In-plane shear strength S

- Shear strain is carried mainly by the matrix: **strain magnification** ($\gamma_m = \gamma_{12}/[1 - V_f(1 - G_m/G_f)]$) leads to matrix yielding / cracking at 45° to the fibres.
- S is of the same order as $Y_t \cdot 2$ (typ. 60-100 MPa) and is strongly non-linear (shear stress-strain curve softens).

II. Anisotropic Strength and Failure Theories

Because composite strength is **directional**, isotropic criteria (Tresca, von Mises) fail. Anisotropic criteria use the **five basic strengths** and are applied in the **material axes**.

2.1 Variation of strength with fibre angle

A lamina loaded in tension at angle θ to the fibres has strength $\sigma_x(\theta)$:

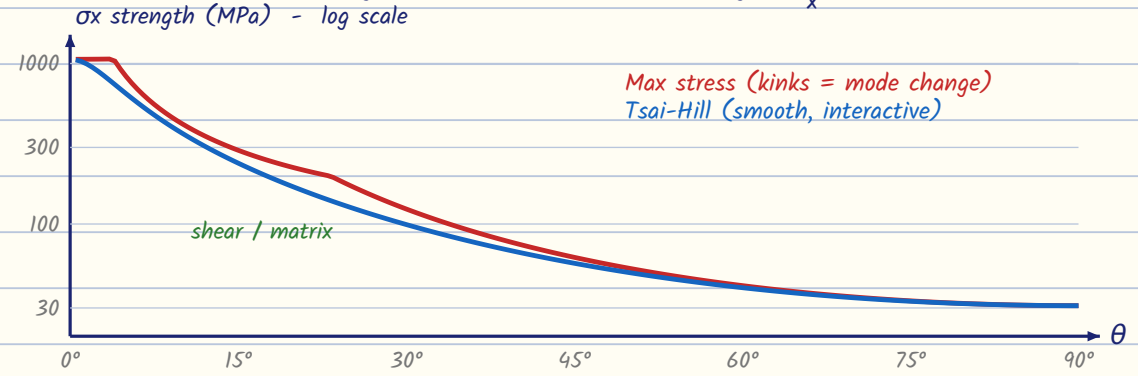


Fig. 2 Off-axis tensile strength of glass/epoxy vs fibre angle

- A small misalignment (≈ 5-10°) cuts the strength by **60-65 %**: shear and transverse stresses appear immediately (at 10°: $\sigma_1 = 0.97\sigma_x$, $\sigma_2 = 0.03\sigma_x$, $\tau = 0.17\sigma_x$) and the weak S and Y are exceeded.
- Near 0° → fibre-controlled; 10°-40° → **shear-controlled**; near 90° → transverse (matrix) controlled.
- The **max stress** theory gives a jagged curve; **Tsai-Hill** / **Tsai-Wu** give a smooth curve that matches test data (interaction).

2.2 Summary of criteria

Criterion	Equation (plane stress)
Max stress	$\sigma_1 = X$, $\sigma_2 = Y$, $\lvert\tau_{12}\rvert = S$ (any one)
Max strain	$\varepsilon_1 = X/E_1$, $\varepsilon_2 = Y/E_2$, $\gamma_{12} = S/G_{12}$

Criterion	Equation (plane stress)
Hill / Tsai-Hill	$\sigma_1^2/X^2 - \sigma_1\sigma_2/X^2 + \sigma_2^2/Y^2 + \tau_{12}^2/S^2 = 1$
Hoffman	$\sigma_1^2/(X_tX_c) - \sigma_1\sigma_2/(X_tX_c) + \sigma_2^2/(Y_tY_c) + (X_c - X_t)\sigma_1/(X_tX_c) + (Y_c - Y_t)\sigma_2/(Y_tY_c) + \tau_{12}^2/S^2 = 1$
Tsai-Wu	$F_1\sigma_1 + F_2\sigma_2 + F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau_{12}^2 + 2F_{12}\sigma_1\sigma_2 = 1$
Azzi-Tsai	Tsai-Hill with $\sigma_1\sigma_2$ term replaced by $\lvert\sigma_1\sigma_2\rvert$ (equal tension/compression strengths); used for ±θ laminates
Hashin (mode-based)	Fibre tension: $(\sigma_1/X_t)^2 + (\tau/S)^2 = 1$; matrix tension: $(\sigma_2/Y_t)^2 + (\tau/S)^2 = 1$; fibre compression: $\sigma_1 = -X_c$

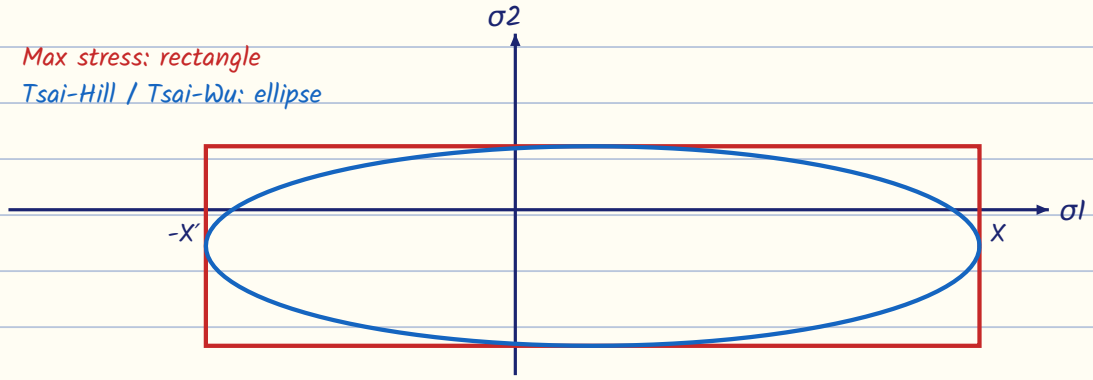


Fig. 3 Failure envelopes for 1-2 (12 = 0): rectangle (max stress) vs ellipse (interactive)

2.3 Hoffman criterion

Hoffman (1967) extended Hill's criterion with **linear terms** to differentiate between tensile and compressive strengths (like Tsai-Wu but with F_{12} fixed = $-\frac{1}{2}F_{11}$). Failure when the LHS = 1. For σ_1 = 500, σ_2 = 20, τ = 30 (glass/epoxy): 0.386 - 0.015 + 0.109 - 0.349 + 0.476 + 0.174 = **0.780 < 1** (safe).

III. Importance of Shear Strength

- The in-plane shear strength S is **small** (≈ 70 MPa for glass / epoxy, 70-100 MPa for carbon / epoxy) and is **matrix-dominated**; it is also non-linear, so shear controls failure for angles 10°-80°.
- **Shear stress** is the main load in ±45° laminates, torsion shafts, thin-wall tubes, joints and webs of beams. In a lamina at θ: $\tau_{12} = -\sigma_x \sin\theta \cos\theta$ - max at 45°.
- At θ = 45° the tension strength is only **57 MPa** (Tsai-Hill) - 5 % of X_t; failure is by shear and transverse tension in the matrix.
- **Interlaminar shear (ILSS)** between plies governs delamination, bolt bearing, impact damage and short-beam failure; typical ILSS of CFRP is 60-90 MPa.

- Shear strength is needed for each failure theory (S appears in all: S^2, $F_{66} = 1/S^2$) so it **must be measured accurately**.

Shear test	Principle	Remark
±45° tension test (ASTM D3518)	Tension on $[\pm 45]_s$ coupon: $S = P / 2A$	Simple; gives in-plane shear curve; large strain
Iosipescu V-notch (ASTM D5379)	Fixture loads a V-notched beam in pure shear	Uniform shear in test section; accurate G_{12}, S
Rail shear (2 or 3 rail)	Plate shear between steel rails	Good, but large specimen
Torsion of thin tube	Pure shear in a cylindrical wall	Best state of stress; expensive
Short-beam shear (ASTM D2344)	3-point bend with small span/depth ratio → interlaminar shear	Quality control of ILSS

IV. Choice of Failure Criteria

Criterion	Advantage	Disadvantage	Typical use
Max stress	Simple; gives failure mode; uses simple tests	No interaction; unconservative; kinks	Preliminary design, mode identification
Max strain	Includes Poisson effect; mode known	No interaction of stresses	Design with strain allowables (aerospace)
Tsai-Hill	Smooth; interaction; one equation	Equal tension and compression assumed; no mode	Ductile-matrix, UD laminae
Tsai-Wu	General; tension ≠ compression; invariant	F_{12} uncertain; no mode information	FEM codes, general laminate analysis
Hashin / Puck / LaRC	Mode-based with physical basis; better for compression	More inputs, complex	Advanced / progressive failure analysis

Guidelines for choosing

> Use **max stress / max strain** when failure **mode** is needed or for quick checks; use **Tsai-Wu / Tsai-Hill** for strength-ratio and envelopes.

> For brittle compression failure prefer criteria with different X_t, X_c (Tsai-Wu, Hoffman, Hashin); for ±θ laminates with equal strengths Tsai-Hill is adequate.

> Always combine with **ply-discount (progressive failure)** and **knock-down factors** (environment, fatigue, scatter, safety factor 1.5-2) for design.

> Compare with test results (World-Wide Failure Exercise: Puck, Tsai-Wu, Zinoviev, Cuntze perform best); **no single criterion is universally best**.

V. Examples

Example 1

Glass/epoxy (X_t = 1062, X_c = 610, Y_t = 31, Y_c = 118, S = 72 MPa). A lamina with fibres at 45° is loaded in uniaxial **compression** σ_x = -50 MPa. Check failure by Tsai-Hill and Tsai-Wu.

$\sigma_1 = \sigma_x \cos^2 45° = -25$, $\sigma_2 = \sigma_x \sin^2 45° = -25$, $\tau_{12} = -\sigma_x \sin 45° \cos 45° = +25$ MPa.

Tsai-Hill (compressive strengths X_c, Y_c): $(25/610)^2 - (25 \times 25)/610^2 + (25/118)^2 + (25/72)^2 = 0.0017 - 0.0017 + 0.0449 + 0.1206 = 0.165 < 1$ → safe; $R = 1/\sqrt{0.165} = 2.46$.

Tsai-Wu: $a = F_{ij}\sigma_i\sigma_j = 0.2795$, $b = F_i\sigma_i = -0.577$ → $a + b = -0.298 < 1$ → safe; $R = (0.577 + \sqrt{(0.333 + 1.118)})/0.559 = 3.19$.

For 45° **tension** the Tsai-Wu strength is only **56 MPa**: compressive loading is much safer here because $Y_c \gg Y_t$.

Example 2

Compare the strength of a glass/epoxy lamina at θ = 10°, 30°, 45°, 60° under uniaxial tension.

θ	10°	30°	45°	60°	90°
Max stress (MPa)	421	124	62	41.3	31
Tsai-Hill (MPa)	367.7	99.2	56.9	40.1	31
Failure mode (max stress)	Shear	Transverse	Shear	Transverse	Transverse

Strength falls from **1062 MPa at 0°** to < 40 % at 10° - the importance of fibre alignment and of shear.

Example 3

A lamina carries σ_1 = 600, σ_2 = 15, τ_{12} = 50 MPa. Does Tsai-Hill predict failure?

$(600/1062)^2 - 600 \times 15/1062^2 + (15/31)^2 + (50/72)^2 = 0.3192 - 0.0080 + 0.2341 + 0.4823 =$ **1.028 > 1** → **fails** (shear + transverse interaction), although max stress says safe (all components below strength: 600 < 1062, 15 < 31, 50 < 72). → max stress is **unconservative** here.

VI. Fibre-Matrix Load Transfer and Failure of Short-Fibre Lamina

When a fibre breaks (or ends), the load is **re-introduced** into it by interfacial shear stress τ_i. The length needed for the fibre stress to reach σ_{fu} is the **critical length** l_c; fibres shorter than l_c **pull out** instead of breaking.

$$l_c = \sigma_{fu}\, d / (2\,\tau_i) \quad \text{(force balance: } \sigma_{fu}(\pi d^2/4) = \tau_i(\pi\, d)(l_c/2))$$

$$\sigma_{lu} = \sigma_{fu} V_f (1 - l_c/2l) + \sigma'_m V_m \;\; (l > l_c) \qquad \sigma_{lu} = \tau_i (l/d) V_f + \sigma'_m V_m \;\; (l < l_c)$$

Example

- $d = 10\ \mu m$, $\sigma_{fu} = 3500$ MPa, $\tau_i = 50$ MPa → $l_c = 3500 \times 10/(2 \times 50) =$ **350 µm** ($l_c/d = 35$).
- For $l = 2$ mm: efficiency $= 1 - 350/(2 \times 2000) =$ **0.91** - short fibres with $l > 10\, l_c$ are almost as effective as continuous fibres.

6.1 Toughening mechanisms (energy absorption)

- **Fibre fracture** and **fibre pull-out** (work $\propto \tau_i\, l_c^2/d$): the main source of toughness in brittle-matrix composites.
- **Debonding** of the interface and **matrix cracking**, crack deflection / bridging by fibres.
- A **weak** interface improves toughness (crack deflection, pull-out) but reduces transverse and shear strength; a **strong** interface raises Y_t, S but gives brittle failure → compromise via fibre sizing / coupling agents.

VII. Failure of Laminates

- **First-ply failure (FPF)**: usually transverse cracking in the off-axis / 90° plies (Y_t, S exceeded).
- **Progressive failure**: failed plies are degraded (ply-discount), then loads are redistributed until the **last-ply failure (LPF)** by fibre fracture.
- **Delamination**: separation of plies by interlaminar stresses σ_z, τ_{xz}, τ_{yz}, which are large at **free edges**, holes, notches and under impact. Resisted by interlaminar toughness G_{Ic}, G_{IIc} (DCB and ENF tests).
- **Fatigue**: matrix cracks → delamination → fibre breaks (stiffness degrades gradually, no sudden failure as in metals); **impact (BVID)**: barely visible internal damage cuts compression strength by 30-50 % (CAI).

- **Environment**: moisture absorption and temperature lower matrix-dominated strengths (Y, S, X_c) - design with "hot-wet" allowables.

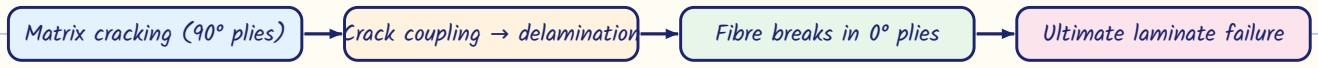


Fig. 4 Sequence of damage in a tensile cross-ply laminate

Damage mode	Cause	Remedy
Matrix cracking	Low Y_t, thermal stress	Toughened resins, ply thickness / angle design
Delamination	Interlaminar stress, impact	Stitching, z-pinning, interleaves, avoid free-edge
Fibre microbuckling	Compression with misalignment	Better fibre alignment, higher G_m, 0° ply support
Fibre breakage	Over-tension	Higher V_f, higher-strength fibre

VIII. More Solved Examples

Example 4

Glass/epoxy lamina with fibres at 30° is loaded by σ_x = 60 MPa tension. Find strength ratio by max stress, Tsai-Hill and Tsai-Wu.

$\sigma_1 = 0.75 \times 60 = 45$, $\sigma_2 = 0.25 \times 60 = 15$, $\tau_{12} = -0.433 \times 60 = -25.98$ MPa.

Max stress: $R = \min(1062/45, 31/15, 72/25.98) = \min(23.6, 2.07, 2.77) =$ **2.07** (matrix transverse tension).

Tsai-Hill: $(45/1062)^2 - (45 \times 15)/1062^2 + (15/31)^2 + (25.98/72)^2 = 0.0018 - 0.0006 + 0.2341 + 0.1302 = 0.3655 \rightarrow R =$ **1.65**.

Tsai-Wu: $a = 0.181$, $b = 0.325$, $a + b = 0.506 \rightarrow R = [-0.325 + \sqrt{(0.1056 + 0.7239)}]/0.362 =$ **1.62**. Failure stress: $\sigma_x = 60 \times R =$ **124 / 99 / 97 MPa** (max stress / Tsai-Hill / Tsai-Wu).

Example 5

Carbon/epoxy with E_1 = 181 GPa, X_t = 1500 MPa: find the ultimate strain along fibres and the stress in a 0° lamina at 0.5 % strain; will it fail?

$\varepsilon_{1t} = X_t/E_1 = 1500/181000 =$ **0.83 %**. At ε_1 = 0.5 %, $\sigma_1 = E_1\varepsilon_1 = 181 \times 0.005 =$ **905 MPa < 1500** → safe, $R = 0.83/0.5 =$ **1.66**.

Example 6

For σ_1 = 500, σ_2 = 20, τ_{12} = 30 MPa (glass/epoxy) apply Hashin matrix-tension criterion.

$(\sigma_2/Y_t)^2 + (\tau_{12}/S)^2 = (20/31)^2 + (30/72)^2 = 0.416 + 0.174 = \mathbf{0.590 < 1}$ → no matrix failure; fibre tension: $(500/1062)^2 + (30/72)^2 = 0.222 + 0.174 = 0.396 < 1$ ✓.

IX. Derivations Worth Knowing

9.1 Strength of off-axis lamina from Tsai-Hill

Put $\sigma_1 = \sigma_x m^2$, $\sigma_2 = \sigma_x n^2$, $\tau_{12} = -\sigma_x mn$ ($m = \cos\theta$, $n = \sin\theta$) in the Tsai-Hill equation:

$$\sigma_x^2[\, m^4/X^2 - m^2n^2/X^2 + n^4/Y^2 + m^2n^2/S^2 \,] = 1$$

$$\mathbf{1/\sigma_x^2 = m^4/X^2 + (1/S^2 - 1/X^2)\, m^2n^2 + n^4/Y^2}$$

At $\theta = 0$: $\sigma_x = X$; at $\theta = 90°$: $\sigma_x = Y$; in between the shear term $(1/S^2)$ makes the curve fall very steeply.

9.2 Tsai-Wu strength ratio

Scale the applied stresses by R ($\sigma \to R\sigma$) and put into the Tsai-Wu equation: $\mathbf{aR^2 + bR - 1 = 0}$, with $a = F_{ij}\sigma_i\sigma_j$ and $b = F_i\sigma_i$. Positive root: $R = [-b + \sqrt{(b^2 + 4a)}] \,/\, 2a$; if $a = 0 \to R = 1/b$. Failure stresses = R × applied stresses.

9.3 Strain magnification factor (transverse loading)

Under σ_2 the lamina strain $\varepsilon_2 = \sigma_2/E_2$ is shared: $\varepsilon_2 h = \varepsilon_f V_f h + \varepsilon_m V_m h$ with $\varepsilon_f = \varepsilon_m E_m/E_f$. Hence

$$\varepsilon_m / \varepsilon_2 = 1 / [1 - V_f(1 - E_m/E_f)]$$

For glass/epoxy $V_f = 0.6$: $S_m = 1/[1 - 0.6(1 - 3.5/72)] = \mathbf{2.3}$; with epoxy fail-strain 4 % → $\varepsilon_{2u} \approx$ 1.7 % at best, and in practice 0.4-0.5 % because of voids and interface flaws.

9.4 Failure envelope in 1-12 plane

For $\sigma_2 = 0$, Tsai-Hill reduces to $(\sigma_1/X)^2 + (\tau/S)^2 = 1$: an **ellipse** with semi-axes X and S; max stress gives a **rectangle**. Interaction shrinks the safe area at the corners.

Quantity	Glass/epoxy	Carbon/epoxy (T300)	Kevlar/epoxy
X_t (MPa)	1062	1500	1400
X_c (MPa)	610	1500	235
Y_t (MPa)	31	40	12
Y_c (MPa)	118	246	53
S (MPa)	72	68	34
X_t/Y_t	34	38	117

Observe: Aramid has very low X_c (kink bands in fibre) - never use Kevlar where compression governs.

X. Design Practice Against Failure

- Orient plies in **several directions** ($[0/\pm45/90]_s$) so that no ply sees large Y or S stresses.
- Keep at least **10 % of plies in each of 0°, ±45°, 90°** (rule of thumb) for damage tolerance.
- Avoid **free-edge** stacks with many same-angle plies together (< 4 adjacent plies); use ply drop-off tapers, stitching or edge caps.
- Use **strain allowables** (0.3-0.4 % for damage-tolerant CFRP) rather than full ultimate strain; apply **safety factor 1.5** on ultimate.
- Test: coupons → elements → sub-components → full scale (**building-block approach**).

XI. Quick Revision - Unit V

Remember

> X_t: fibre fracture; X_c: microbuckling ($G_m/(1 - V_f)$); Y_t, S: matrix / interface; Y_c: matrix shear at 45°.

> Y_t and S are tiny relative to X_t → **first failure is matrix cracking**; angle misalignment of 5-10° drastically lowers strength.

> Max stress & strain (non-interactive, mode known); Tsai-Hill, Hoffman, Tsai-Wu (interactive); Hashin / Puck (mode-based).

> Shear strength tests: ±45° tension, Iosipescu, rail shear, short-beam shear.

Likely exam questions

> Explain micromechanics of failure of a unidirectional lamina in tension, compression and shear.

> Discuss anisotropic strength and failure theories; compare Tsai-Hill, Hoffman and Tsai-Wu.

> Why is shear strength important? How is it measured?

> How do you choose a failure criterion? Solve examples.

Short Answer Questions (2 marks)

Q1. What is the main failure mode in longitudinal compression?

Ans: Microbuckling (kinking) of fibres.

Q2. Why is Y_t so low?

Ans: Matrix strain magnification, interface debonding and voids.

Q3. Write Rosen's shear-mode buckling stress.

Ans: $\sigma_c = G_m/(1 - V_f)$.

Q4. Define critical fibre length.

Ans: $l_c = \sigma_{fu} d/(2\tau_i)$: the minimum length for fibre stress to reach its strength.

Q5. How does misalignment affect strength?

Ans: Introduces shear and transverse stresses, lowering strength sharply (≈ 60 % at 10°).

Q6. Name two shear test methods.

Ans: ±45° tension test, Iosipescu V-notch test (also rail shear).

Q7. Why is Tsai-Wu preferred in FEM?

Ans: It is a single smooth invariant criterion with tension-compression differences and interaction.

Q8. Which criterion gives mode of failure?

Ans: Maximum stress / strain and Hashin / Puck criteria.

Q9. What is ply discount?

Ans: Reducing the stiffness of failed plies in progressive failure analysis.

Glossary of Key Terms

Term	Meaning
Microbuckling	Local instability of fibres in compression, forming kink bands
Ineffective length	Length of broken fibre over which load is not fully carried
Strength ratio R	Failure load / applied load (R > 1 safe)
Free-edge effect	High interlaminar stresses near laminate edges causing delamination
ILSS	Interlaminar shear strength from short-beam test
CAI	Compression after impact strength
BVID	Barely visible impact damage
Progressive failure	Sequential ply failure with stiffness degradation

Answer Outlines for 10-mark Questions

Micromechanics of failure of a UD lamina

> Sketch of 5 failure modes.

> Tension: fibre strength, V_{min}, V_{crit}; compression: Rosen modes; transverse tension / compression; shear.

Compare failure theories

> Max stress, max strain, Tsai-Hill, Hoffman, Tsai-Wu equations.

> Envelope sketch, interaction, tension ≠ compression, mode information.

Importance of shear strength

- Low S, matrix dominated; off-axis loading; torsion, ±45° laminates.
- Tests: ±45°, Iosipescu, rail shear; ILSS.